FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Birth Asphyxia & Hypoxic-Ischaemic Encephalopathy

Tutorial | Week 4 · Small babies, sugar & the heart | Neonatology | Dr Namugga

Birth asphyxia is failure to start or sustain breathing at birth because of interrupted oxygen delivery around labour; the brain injury that follows is hypoxic-ischaemic encephalopathy (HIE). Grade it with Sarnat staging, give meticulous supportive care (temperature, glucose, fluids, ventilation, seizure control with phenobarbital) and avoid secondary injury; in Uganda therapeutic cooling is not standard care outside tertiary NICUs.

Must know before the ward round

1. Birth asphyxia (WHO) - failure to initiate and sustain breathing at birth. HIE is the brain injury that follows; neonatal encephalopathy is the clinical syndrome (altered consciousness, tone, feeding, seizures).
2. Supporting criteria for an intrapartum hypoxic cause - cord/early arterial pH below 7.0 or base deficit 12 mmol/L or more, Apgar 0-3 beyond 5 min (or below 5 at 5 and 10 min), early encephalopathy, multi-organ dysfunction, a sentinel event.
3. Sarnat - mild (hyperalert, normal/increased tone, no seizures, usually normal outcome); moderate (lethargic, hypotonic, weak suck, seizures common); severe (stupor/coma, flaccid, absent reflexes, high death and disability).
4. Two waves of injury - primary energy failure at the time of the insult, then a latent phase of about 6 h, then secondary energy failure over 6-72 h. Treatment aims to prevent secondary injury.
5. Temperature 36.5-37.5 C - AVOID hyperthermia (worsens brain injury). Do not passively cool babies in Ugandan wards without a NICU protocol.
6. Therapeutic hypothermia (33.5 C for 72 h, started within 6 h, 36 weeks or more) reduces death/disability in high-income NICUs, but the HELIX trial (India, Sri Lanka, Bangladesh, 2021) showed no benefit and more deaths in LMICs.
7. Glucose - check early and often; treat below 2.6 mmol/L with 10% dextrose 2 mL/kg IV; avoid both hypo- and hyperglycaemia.
8. Fluids - start 10% dextrose 60 mL/kg/day on day 1 (often restricted to 40-60 mL/kg/day if oliguric); monitor urine output (oliguria below 1 mL/kg/h), weight and sodium (SIADH, acute kidney injury).
9. Seizures - phenobarbital 20 mg/kg IV over 15-20 min; extra 10 mg/kg doses to a total of 40 mg/kg; maintenance 5 mg/kg/day once daily. Check glucose and calcium; watch for apnoea.
10. Second-line anticonvulsants - phenytoin 20 mg/kg IV over 20-30 min, or levetiracetam 40-60 mg/kg IV, or midazolam; avoid repeated diazepam in neonates.
11. Prognosis - a baby still floppy or spastic, unresponsive and unable to suck at 1 week has severe brain injury and will do poorly (WHO); recovery of suck and normal tone by 1 week is reassuring.
12. Follow up every survivor of moderate/severe HIE for cerebral palsy, epilepsy, developmental delay, hearing and vision problems, and feeding difficulty.

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1. Definition

Birth asphyxia means the baby's oxygen supply was interrupted before, during or just after birth so that he fails to start or sustain breathing. The resulting brain damage is **hypoxic-ischaemic encephalopathy (HIE)**. "Neonatal encephalopathy" is the broader clinical term for a term baby with abnormal consciousness, tone, feeding or seizures; HIE is its commonest cause.

Term	Meaning
Birth asphyxia (WHO)	Failure to initiate and sustain breathing at birth
Perinatal asphyxia (physiological)	Impaired gas exchange → **hypoxaemia, hypercapnia and metabolic acidosis**
Neonatal encephalopathy (NE)	Clinical syndrome in a baby of ≥35 weeks: reduced consciousness or seizures, abnormal tone and reflexes, poor feeding, often breathing difficulty
HIE	NE caused by intrapartum hypoxia-ischaemia
Apgar score	0–10 at 1 and 5 min; describes the baby's state, **does not diagnose asphyxia** and is never used to decide whether to resuscitate

Criteria that support an intrapartum hypoxic-ischaemic cause (ACOG/AAP "Neonatal encephalopathy and neurologic outcome"; used in research and medicolegal work):

1. **Cord (or early) arterial pH <7.0** and/or **base deficit ≥12 mmol/L**.
2. **Apgar 0–3 beyond 5 minutes** (or <5 at 5 and 10 minutes).
3. Early-onset **moderate or severe encephalopathy**.
4. **Multi-organ dysfunction** (kidney, liver, heart, gut, blood).
5. A **sentinel event** (cord prolapse, abruption, uterine rupture, shoulder dystocia) and fetal heart abnormalities.
6. Neuroimaging consistent with acute hypoxic injury (basal ganglia and thalamus, watershed).

UGANDA CONTEXT

Cord blood gases are rarely available in Ugandan public hospitals. On the ward, asphyxia is diagnosed **clinically**: a history of a difficult or prolonged labour or fetal distress, a baby who needed resuscitation (low Apgar), and **encephalopathy** in the first hours to days. Write all three in your notes.

CLINICAL PEARL

The one thing never to forget: **prevention is the treatment.** A well-monitored labour (partograph, fetal heart every 30 min in active labour, timely action) and a trained person with a working bag and mask at every birth prevent most HIE.

2. Epidemiology

- **Global**: intrapartum-related events cause roughly a quarter of neonatal deaths and over a million stillbirths a year, almost all in low- and middle-income countries (LMICs) Verify *figures against the latest WHO/UNICEF estimates*.
- **Uganda**: intrapartum-related events (asphyxia) are one of the **three leading causes of neonatal death** (with prematurity and infection), a major cause of **fresh stillbirth**, and the commonest preventable cause of **cerebral palsy**.
- Many asphyxiated babies in Uganda come from **prolonged or obstructed labour**, delayed referral from lower-level facilities, home births, inadequate fetal monitoring and a lack of skilled resuscitation.

Services by level of care Verify *compiled from the source notes, not an MoH equipment list*

Level	Usually available	Often missing
HC III / HC IV	HBB resuscitation (bag and mask), phenobarbital injection, ampicillin and gentamicin, glucometer at some	Oxygen and pulse oximetry at HC III, IV infusions for neonates, SCBU
General Hospital / RRH	Newborn unit (SCBU), oxygen, IV fluids, phenobarbital, glucose monitoring, NGT feeding	Electrolytes at some, CPAP at some, blood gas
NRH and tertiary (Mulago/Kawempe, Mengo, Nsambya, Lubaga)	CPAP and ventilation, blood gases, electrolytes, cranial ultrasound	**Therapeutic hypothermia** is not standard in public Ugandan units; **aEEG/EEG and MRI** in only a few Kampala centres

3. Aetiology and Risk Factors

Most cases follow prolonged or obstructed labour, poorly monitored labour, cord prolapse, abruption, ruptured uterus or delayed resuscitation.

Antepartum and maternal	Intrapartum	Fetal / neonatal
Eclampsia, maternal shock	**Prolonged second stage, obstructed labour**	Preterm and post-term babies
APH (abruption)	Malpresentation, breech delivery	Twin delivery
Maternal infection (chorioamnionitis)	**Cord prolapse**	Thick meconium
Lack of ANC, home birth	**Ruptured uterus**	Shoulder dystocia
Delayed referral	Oxytocin overuse (**hyperstimulation**)	Delayed or ineffective resuscitation
	Inadequate fetal heart monitoring	

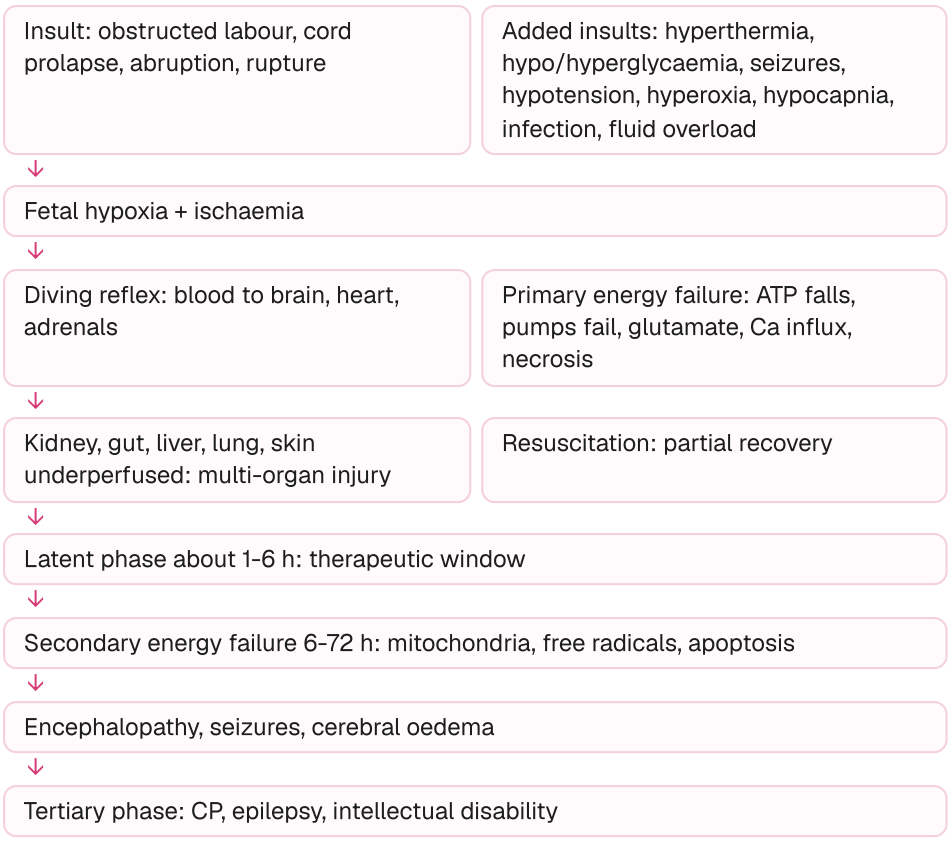
4. Pathophysiology

The two phases of brain injury

1. **Insult (hypoxia + ischaemia)**.
2. **Primary energy failure**: ATP falls, Na^+/K^+ pumps fail, cytotoxic oedema, glutamate release, calcium influx, cell necrosis.
3. **Resuscitation** → partial recovery of brain energy.
4. **Latent phase** (about 1–6 hours): the "therapeutic window" (cooling must start here).
5. **Secondary energy failure** (6–72 h): mitochondrial failure, free radicals, inflammation, apoptosis, seizures.

6. **Tertiary phase** (weeks to months): ongoing injury, impaired repair.

Everything that adds to the energy deficit in the secondary phase makes the injury worse: **hyperthermia, hypoglycaemia, hyperglycaemia, seizures, hypotension, hypoxaemia, hyperoxia, hypocapnia (over-ventilation reduces cerebral blood flow), infection and fluid overload**.



The "diving reflex" and multi-organ injury

When oxygen falls, the fetus redistributes blood to the **brain, heart and adrenals** at the expense of the **kidneys, gut, liver, skin and lungs**. So asphyxia damages several organs:

Organ	Consequence	What to monitor
Brain	HIE, seizures, cerebral oedema	Neurological examination, seizures
Kidneys	Acute tubular necrosis → **oliguria, AKI**, hyponatraemia	Urine output, creatinine, sodium
Heart	Myocardial dysfunction, tricuspid regurgitation, hypotension	HR, CRT, BP
Lungs	Meconium aspiration, persistent pulmonary hypertension (PPHN), haemorrhage	SpO_2 (pre- and post-ductal)
Liver	Raised transaminases, coagulopathy, hypoglycaemia	Glucose, bleeding
Gut	Feed intolerance, NEC	Abdomen, aspirates
Blood	Thrombocytopenia, **DIC**	Bleeding, platelets
Endocrine / metabolic	**Hypoglycaemia**, hypocalcaemia, SIADH	Glucose, calcium, sodium

Patterns of brain injury

- **Acute profound ("near total")** insult (for example cord prolapse, uterine rupture): deep grey matter (**basal ganglia, thalami**) → dyskinetic or spastic quadriplegic CP.

- **Prolonged partial** insult (for example long obstructed labour): **watershed** cortex and white matter → cognitive problems, spastic CP, cortical visual impairment.

Feature links

- Cortical and brainstem energy failure → **altered consciousness** (hyperalert → lethargy → coma).
- Early sympathetic surge (mild) → **dilated pupils, tachycardia, jitteriness**.
- Parasympathetic dominance (moderate) → **constricted pupils, bradycardia, secretions**.
- Brainstem injury (severe) → **apnoea, absent suck and Moro, poor pupil responses**.
- Neuronal hyperexcitability (glutamate, calcium) → **seizures**, often subtle, typically starting within 12–24 h.
- Cortical and basal ganglia injury → **hypotonia** early, later **spasticity or dyskinesia**.
- Acute tubular necrosis → **oliguria**, rising creatinine and potassium; ADH excess → **SIADH, hyponatraemia, weight gain**.
- Myocardial ischaemia → **hypotension, poor perfusion**, tricuspid regurgitation murmur.
- Hypoxic pulmonary vasoconstriction → **PPHN**: labile SpO_2, pre- and post-ductal difference.
- Gasping in utero → **meconium aspiration**.
- Hepatic injury → **coagulopathy, hypoglycaemia**; endothelial injury → **DIC, bleeding**.
- Gut ischaemia → **feed intolerance, NEC**.

5. Classification / Staging

Sarnat staging (modified): the ward version

Feature	Stage 1 (mild)	Stage 2 (moderate)	Stage 3 (severe)
Level of consciousness	**Hyperalert**, irritable, jittery	**Lethargic** or obtunded	**Stupor or coma**
Tone	Normal or increased	**Mild hypotonia**	**Flaccid**
Posture	Mild distal flexion	Strong distal flexion	Intermittent decerebration
Stretch reflexes	Brisk	Brisk	Decreased or absent
Moro	Strong, low threshold	Weak, incomplete	Absent
Suck	Weak	Weak or absent	Absent
Autonomic	Sympathetic: **dilated pupils**, tachycardia	Parasympathetic: **constricted pupils**, bradycardia, secretions	Both depressed: pupils variable, poor light reflex, apnoea
Seizures	**None**	**Common** (often subtle)	Uncommon clinically (EEG suppressed) or frequent early
Duration	<24 h	2–14 days	Hours to weeks
Outcome	**Usually normal**	**Variable**: about 20–40% die or have disability (lower with cooling in high-income countries)	**Death in about half or more; nearly all survivors disabled**

EXAM FAVOURITE

Examiners love: "Mild = hyperalert + no seizures + normal outcome; moderate = lethargy + hypotonia + seizures; severe = coma + flaccid." Also: **pupils dilated in mild (sympathetic), constricted in moderate (parasympathetic)**.

Thompson score (0–22; tone, consciousness, fits, posture, Moro, grasp, suck, respiration, fontanelle) is a simple bedside alternative, used in African and Asian research and some Ugandan units. A higher score means worse encephalopathy; scores of about 15 or more on days 1–7 predict poor outcome.

6. Clinical Features

History

Question	Why it matters
Labour: duration, augmentation, partograph, obstructed, malpresentation	Mechanism and severity of the insult
Fetal heart: bradycardia, decelerations, meconium	Evidence of fetal hypoxia
Sentinel event: cord prolapse, abruption, rupture, eclamptic fit, shoulder dystocia	Supports an intrapartum cause, predicts pattern
Resuscitation: time to first breath or cry, bag-mask duration, HR response, compressions or adrenaline, Apgar 1/5/10	Longer resuscitation = more severe HIE
Gestation and birth weight	Cooling eligibility (≥36 weeks); preterm encephalopathy has different causes
Maternal fever, ROM duration (>18 h), foul liquor	**Sepsis or meningitis** can mimic or add to HIE
Maternal drugs (opioids, $MgSO_4$, diazepam)	Can cause depression without HIE
Seizures noticed: when, what they looked like	Timing (HIE seizures typically start within 12–24 h)
Feeding, urine and meconium passed since birth	Suck = neurological status; oliguria = AKI
Maternal HIV (last viral load), syphilis, blood group	Prophylaxis and other causes
Immunisation (BCG, OPV0)	Give before discharge

Examination

Follow Hutchison's Clinical Methods (24th edition) order adapted for the newborn. Resuscitation and stabilisation come first; then a full examination with the **Sarnat items recorded one by one**. **Repeat the Sarnat examination at least daily** (and at 6 h if cooling were possible): encephalopathy evolves.

Introduction, consent and preparation

- Introduce yourself to the mother; confirm the baby's name, age in hours and sex; explain and gain consent.
- **Hand hygiene**.
- **Position and exposure**: under a radiant warmer or in a warm room, fully undressed only for as long as needed; watch the temperature (do not overheat).

General inspection (before touching)

Sign	Mechanism	What it means
Hyperalert, jittery, wide-eyed	Sympathetic surge	Sarnat 1
Lethargic, little spontaneous movement	Cortical depression	Sarnat 2
Stupor or coma, no spontaneous movement	Severe cortical and brainstem injury	Sarnat 3
Posture: distal flexion; decerebrate posturing	Tone abnormality	Stage 2 (strong flexion) or 3 (decerebration)

Sign	Mechanism	What it means
Subtle seizures: lip smacking, eye deviation, cycling, apnoea; clonic, tonic, myoclonic jerks	Neuronal hyperexcitability	Seizures: treat
Colour: cyanosis, pallor, plethora, jaundice; meconium staining	Hypoxaemia, PPHN; blood loss; polycythaemia	Other organ involvement
Breathing pattern: irregular, apnoea, recession	Brainstem injury, meconium aspiration	Respiratory support
Dysmorphism	Syndrome or malformation	Other cause of encephalopathy

Vital signs and growth: temperature (avoid >37.5 °C), HR, RR and apnoea, SpO_2 (pre- and post-ductal if PPHN is suspected), CRT (<3 s), BP if available; weight (and daily weights), length and **head circumference** plotted.

Hands, face and neck

- **Hands**: grasp; peripheral perfusion; fisting.
- **Face**: pupils (dilated in mild, constricted in moderate, poor light reflex in severe); suck (finger in the mouth); facial movement.
- **Head**: fontanelle (bulging = meningitis or cerebral oedema), sutures, scalp swelling: **caput** (crosses sutures, soft, present at birth) versus **cephalhaematoma** (limited by sutures) versus **subgaleal haemorrhage** (boggy, crosses sutures, spreads to the neck and ears, falling Hb, shock).

Nervous system (Sarnat items)

- **Level of consciousness and spontaneous activity**.
- **Tone**: pull-to-sit (head lag), ventral suspension, limb recoil.
- **Posture**; **stretch reflexes**; **Moro**; **grasp**; **suck**.
- **Autonomic**: pupils, HR, secretions.
- **Seizures**: describe and time them.

Completing the examination

- **Chest**: meconium aspiration (crackles, recession), PPHN.
- **Heart**: murmur of tricuspid regurgitation, poor perfusion.
- **Abdomen**: liver size, distension (NEC).
- **Skin and bleeding**: puncture sites, cord, gut; jaundice; skin lesions.
- **Urine output**: weigh nappies.
- **Thank** the mother, re-dress the baby, **hand hygiene**, **summarise** with the Sarnat stage and time.
- Document key negatives: "no bulging fontanelle, no dysmorphism, no subgaleal swelling".

Red flags

RED FLAG

- **Apnoea** or irregular breathing, SpO_2 falling
- **Seizures**, especially repeated or prolonged; seizure in the first hours
- **Coma, flaccidity, absent suck and Moro** (severe HIE)
- **Hypotension**, poor perfusion, bradycardia
- **No urine for 24 h**, rising weight, oedema (AKI, fluid overload)
- **Bleeding** from puncture sites, gut or cord (DIC)
- **Hypoglycaemia** (<2.6 mmol/L) or hyperthermia (>37.5 °C)
- **Bulging fontanelle** or fever (think meningitis)

Thinking like a clinician

- **Not every floppy baby after a hard labour has HIE.** Always consider **hypoglycaemia, sepsis and meningitis**, maternal drugs, intracranial haemorrhage (subgaleal, subdural after instrumental delivery), metabolic disease, congenital neuromuscular disorders.
- **Encephalopathy without a hypoxic history** (normal labour, good Apgar) points away from HIE: look for infection, metabolic disease, stroke, or malformation.
- **Seizures in HIE are often subtle**: a baby with apnoea and eye deviation is seizing until proven otherwise.
- **Over-treatment harms too**: hyperthermia from radiant heaters, hyperoxia from 100% oxygen, fluid overload from "routine" 60–90 mL/kg/day in an oliguric baby, and stacked anticonvulsants causing apnoea.
- **Mild HIE today can be moderate tomorrow.** Re-examine; write the stage and the time.
- **Prognosis needs time.** Avoid firm predictions on day 1; recovery of suck and tone over the first week is the best clinical guide in Uganda.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Neonatal sepsis / meningitis	Maternal fever, ROM >18 h, fever or hypothermia, bulging fontanelle, onset after a normal period	Blood culture, CRP, LP
Hypoglycaemia	Jittery, lethargic, seizures; SGA or infant of a diabetic mother; responds to glucose	Glucose
Intracranial / subgaleal haemorrhage	Instrumental delivery, boggy scalp crossing sutures, falling Hb, shock	Hb, head circumference, cranial ultrasound
Maternal drug effect	Opioid or $MgSO_4$ before delivery; depression improves over hours; no seizures	History; naloxone not routine
Inborn error of metabolism	Normal at birth, then lethargy, vomiting, seizures after feeds; consanguinity	Glucose, ammonia, lactate, ketones
Neonatal stroke	Focal seizures in a well-looking term baby, often day 1–3	Cranial ultrasound, MRI
Congenital neuromuscular disease / malformation	Hypotonia from birth without encephalopathy; dysmorphism, contractures	Genetics, imaging
Hypocalcaemia / hypomagnesaemia	Jitteriness, seizures, infant of a diabetic mother, preterm	Calcium, magnesium
Kernicterus	Day 3–7, deep jaundice, opisthotonus, high-pitched cry	Bilirubin

HIE itself: hypoxic labour, low Apgar, resuscitation, encephalopathy from birth, multi-organ dysfunction; diagnosed clinically, supported by blood gas and imaging.

8. Clinical Workup

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes*.

Bedside

Test	Why	Expected / abnormal finding	Available at
Glucose	Hypo- and hyperglycaemia both worsen brain injury	<2.6 mmol/L (treat); >10 mmol/L (reduce glucose infusion rate). Check within the first hour and every 3–6 h for 24–48 h	HC III and above (glucometer)
Temperature	Hyperthermia worsens injury	Keep 36.5–37.5 °C; avoid >37.5 °C	All levels
SpO_2	Avoid hypoxaemia and hyperoxia	Target 90–95%; pre- and post-ductal difference >3–5% suggests PPHN	HC IV and above
Urine output (weigh nappies: 1 g = 1 mL)	AKI	**<1 mL/kg/h** after the first 24 h = oliguria	All levels
Daily weight	Fluid balance	Weight gain in the first week = fluid retention	All levels
HemoCue Hb / PCV	Blood loss (subgaleal), polycythaemia	Falling Hb	HC IV and above
20-minute whole-blood clotting test	DIC	No clot at 20 min	All levels with a glass tube
Sarnat (or Thompson) score	Grade and trend	Stage and time recorded	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC, platelets	Bleeding, sepsis	Thrombocytopenia, anaemia	Platelets, transfusion	HC IV and above
Electrolytes, urea, creatinine	Day 1–3	Hyponatraemia (SIADH), rising creatinine and potassium (AKI)	Fluid restriction; stop potassium; adjust gentamicin	Regional and national referral
Calcium, magnesium	Seizures	Hypocalcaemia (total calcium <1.75–2.0 mmol/L)	10% calcium gluconate 2 mL/kg slow IV	Referral hospitals
Blood culture, CRP	Risk factors or signs of sepsis	Positive culture	Antibiotic duration	Referral hospitals
LFTs, coagulation	Severe HIE, bleeding	Raised transaminases; DIC	Drug dosing; FFP, vitamin K	Referral hospitals

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Cranial ultrasound	Day 1–3	Oedema, haemorrhage; excludes other lesions	Prognosis	Kampala tertiary

Test	When	Expected / abnormal finding	How it changes management	Available at
MRI brain	Day 5–14	Basal ganglia and thalamus or watershed injury	**Best prognostic test**	Few private and tertiary centres

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Cord / arterial blood gas	At birth or first hour	pH <7.0, base excess ≥12	Supports intrapartum asphyxia; cooling eligibility in high-income settings	Few Kampala centres
Lumbar puncture	Fever, bulging fontanelle, seizures without clear HIE, positive culture	Meningitis	21 days IV antibiotics	Hospital, once stable
aEEG / EEG	Seizures, grading	Background suppression, electrographic seizures	Seizure treatment, prognosis	Very limited
Ammonia, lactate	Encephalopathy without a hypoxic history	Inborn error of metabolism	Metabolic referral	NRH

Putting it together

Model one-line summary

> "A 14-hour-old term male, birth weight 3.4 kg, born by vacuum after **obstructed labour for 6 hours** with thick meconium and FHR 90/min, needing **4 minutes of bag-mask ventilation** (Apgar 2, 5, 7), now with **moderate HIE (Sarnat 2)**: lethargic, hypotonic, weak suck, constricted pupils, with **two subtle seizures** (lip smacking and eye deviation) treated with phenobarbital. Temperature 37.0 °C, glucose 3.8 mmol/L, SpO_2 95% in air, no urine yet."

Problem list

1. Moderate HIE with seizures.
2. Risk of AKI or SIADH (no urine at 14 h).
3. Possible meconium aspiration.
4. Sepsis to be excluded.
5. Feeding (unable to suck).
6. Family counselling.

Model case presentation (ward round)

> "**Baby Okello**, a **2-day-old** term male, birth weight **3.5 kg**, born at Mengo by emergency CS for **obstructed labour** (referred from a HC IV after 14 hours of labour) with **thick meconium** and **FHR 80/min**. He was **not breathing at birth**, needed **5 minutes of bag-mask ventilation**, HR rose above 100 at 3 minutes; Apgar 2, 4 and 7.
>
> On day 1 he had **moderate HIE (Sarnat 2)**: lethargic, hypotonic, weak suck, constricted pupils, with **three seizures** (cycling movements and eye deviation) controlled with **phenobarbital 20 mg/kg** then a further **10 mg/kg**; now on **5 mg/kg/day**. Temperature 36.8–37.3 °C. He has had no seizures for 20 hours. Glucose 3.5–6.0 mmol/L. On **10% dextrose 60 mL/kg/day**; passed **1.2 mL/kg/h** urine in the last 12 hours. Sodium 132. Blood culture pending; on **ampicillin and gentamicin** (ROM 20 h). Started **EBM 10 mL/kg/day by NGT** today.

> Mother **HIV-negative** in labour, RPR negative. **BCG and OPV0 not yet given** (will give before discharge). Weight today **3.4 kg** (–3%).
>
> **Assessment:** moderate HIE, improving, seizures controlled; problems: feeding, fluid balance, sepsis exclusion, family counselling. **Plan:** daily Sarnat, advance feeds as suck returns, stop antibiotics at 48 h if culture negative and well, continue phenobarbital and review, development follow-up at 6 weeks."

Worked example 1: fluids in an oliguric baby. A 3.0 kg baby with moderate HIE has passed 10 mL of urine in 24 h (0.14 mL/kg/h) and his weight has risen from 3.0 to 3.15 kg. Sodium is 128 mmol/L. **Interpretation**: **oliguria + weight gain + hyponatraemia = fluid retention (AKI or SIADH)**. **Plan**: restrict to **insensible losses (about 30–40 mL/kg/day) plus urine output**, no potassium, recheck electrolytes and creatinine, review gentamicin (prolong interval or stop if culture negative), weigh twice daily.

Worked example 2: phenobarbital. A 3.2 kg baby → loading **64 mg** IV over 20 min; if still seizing after 15–30 min, **32 mg** (10 mg/kg), repeat once if needed (total 40 mg/kg = 128 mg); maintenance **16 mg once daily**.

Worked example 3: glucose infusion rate. GIR = (% dextrose × mL/kg/day) ÷ 144. 10% dextrose at 60 mL/kg/day = 600 ÷ 144 = **4.2 mg/kg/min**.

9. Management

There is no antidote. In Ugandan units, the treatment is **meticulous supportive care**: keep the baby normothermic (and **never overheated**), keep glucose normal, give the right fluids, support breathing, **stop seizures with phenobarbital**, treat infection, feed carefully and follow up.

Immediate / emergency

First 5 minutes

1. **Resuscitation** as per HBB and neonatal resuscitation (see that topic): dry, airway, **bag-mask ventilation by 60 s**, HR checks, compressions and adrenaline in hospital. Ventilation is the key intervention.
2. Once breathing, **admit** if: prolonged ventilation (>1–2 min), abnormal tone, poor suck, seizures, or ongoing respiratory distress.
3. Call the senior; inform and support the mother.

ABCDE

- **A**: position, suction if secretions.
- **B**: oxygen only to keep SpO_2 **90–95%** (avoid hyperoxia); bag-mask for apnoea; CPAP or ventilation for respiratory failure; avoid over-ventilation (hypocapnia).
- **C**: CRT and HR; **IV access** (peripheral or umbilical venous catheter); normal saline **10 mL/kg** over 10–20 min only if clearly hypovolaemic (for example blood loss, subgaleal haemorrhage); dopamine where available for hypotension due to myocardial dysfunction.
- **D**: **glucose** within the first hour; treat <**2.6 mmol/L** with **10% dextrose 2 mL/kg IV**; **seizure?** → **phenobarbital 20 mg/kg IV**.
- **E**: **temperature 36.5–37.5 °C**. Turn the radiant warmer down if >37.5 °C; do not overwrap. **Do not actively cool** without a cooling protocol and NICU; no ad hoc cooling with ice or fans. **Vitamin K** if not given.
- **Sepsis risk** → blood culture, **ampicillin + gentamicin**.

Definitive

Emergency algorithm (first 24 hours)

1. **Resuscitate effectively** (HBB/NRP).

2. **Admit** to the newborn unit (SCBU) every baby with encephalopathy, prolonged resuscitation, or an abnormal examination.
3. **Thermoregulation**: **36.5–37.5 °C**; avoid hyperthermia.
4. **Oxygenation and ventilation**: SpO_2 **90–95%**.
5. **Circulation**: saline only if hypovolaemic; inotropes where available.
6. **Glucose**: keep **≥2.6 mmol/L** (many units target 2.6–8 mmol/L); bolus then infusion.
7. **Fluids**: **10% dextrose 60 mL/kg/day** on day 1; **restrict** (for example 40 mL/kg/day, or insensible losses + urine output) if oliguric or hyponatraemic.
8. **Seizures**: **phenobarbital** (drug table); exclude hypoglycaemia, hypocalcaemia and meningitis.
9. **Antibiotics**: ampicillin + gentamicin if sepsis cannot be excluded; take care with gentamicin in AKI (extend interval, check levels if available).
10. **Feeding**: nil by mouth or small EBM feeds by NGT while unstable; start and advance as the baby improves.
11. **Monitor**: temperature, HR, RR, SpO_2, glucose, urine output, weight, Sarnat stage, seizures.
12. **Communicate** with the family; document carefully.

Drug table

Drug	Dose (per kg; neonate)	Route	Frequency	Duration	Key cautions
Phenobarbital (first line, WHO/UCG)	**Loading 20 mg/kg**; if seizures persist, **further 10 mg/kg** doses up to **40 mg/kg total**; **maintenance 5 mg/kg/day**	IV slowly over **15–20 min** (IM if no IV)	Load once; maintenance once daily (start 12–24 h after load)	Stop when seizure-free (often before discharge; senior decision)	Watch for **apnoea and hypotension**; bag and mask ready
Phenytoin (second line)	**Loading 20 mg/kg**; maintenance 3–5 mg/kg/day	IV in normal saline over **20–30 min** (max 1 mg/kg/min)	Load once; maintenance in 2 doses	Per response	Not in dextrose (precipitates); arrhythmias: ECG or HR monitoring
Levetiracetam (second line where available)	**40–60 mg/kg** loading, then 10–30 mg/kg	IV / oral	Maintenance 12-hourly per local protocol	Per response	Fewer side effects; less effective than phenobarbital as first line (NEOLEV2 trial)
Midazolam (refractory, NICU)	0.05–0.15 mg/kg bolus, then infusion	IV	Infusion	NICU only	Respiratory depression; needs ventilation support
10% calcium gluconate (if hypocalcaemic)	**2 mL/kg**	**Slow IV** over 10 min with HR monitoring	Then oral or IV maintenance	Until calcium normal	Extravasation burns; bradycardia if fast
10% dextrose (if hypoglycaemic)	**2 mL/kg** bolus; infusion 60 mL/kg/day on day 1	IV	Then infusion	Recheck glucose in 30 min	Avoid hyperglycaemia (>10 mmol/L)
Normal saline (only if hypovolaemic)	**10 mL/kg**	IV	Over 10–20 min	Single bolus, reassess	Fluid overload in myocardial dysfunction

Drug	Dose (per kg; neonate)	Route	Frequency	Duration	Key cautions
Dopamine	**5–10 µg/kg/min**	IV infusion	Continuous	Until BP stable	Where available; needs monitoring
Ampicillin + gentamicin	Neonatal doses per the sepsis protocol (common neonatal disorders topic)	IV / IM	Per age and weight	Stop at 48 h if well and culture negative	Extend gentamicin interval in AKI; check levels if available
Vitamin K	1 mg (term); 0.5 mg (preterm) Verify *dose against UCG*	IM	Once	Single dose	If not given at birth

DRUG DOSING

Diazepam is **not** recommended as a routine neonatal anticonvulsant (respiratory depression; the sodium benzoate in it displaces bilirubin from albumin).

Referral criteria

Level	What to do	Refer up when
HC III / HC IV	HBB resuscitation; keep warm (not hot); breastfeed or EBM by cup if able; check glucose or give EBM or sugar water by NGT; **first dose of phenobarbital 20 mg/kg IM** if convulsing; first dose of ampicillin + gentamicin if risk factors	Every baby with encephalopathy, seizures, prolonged resuscitation or inability to suck: **refer** with the mother and a referral note
General Hospital / RRH	SCBU admission, oxygen, IV fluids, glucose monitoring, phenobarbital, antibiotics, NGT feeding, daily Sarnat	Refractory seizures, need for CPAP or ventilation, AKI needing electrolytes, imaging → NRH or tertiary unit
NRH and tertiary (Kampala)	CPAP and ventilation, blood gases, electrolytes, cranial ultrasound, EEG and MRI in some centres, multidisciplinary follow-up	n/a

Monitoring and follow-up

- **Ward monitoring**: temperature, HR, RR, SpO_2, glucose (3–6-hourly for 24–48 h), urine output, weight (daily or twice daily), electrolytes, Sarnat stage daily, seizure chart.
- **Follow-up plan**: weight and head circumference, development at 6 weeks, 3, 6, 9, 12 and 18–24 months; hearing and vision screening; physiotherapy early; nutrition support; epilepsy care; refer to paediatric neurology and cerebral palsy and rehabilitation services (for example Mulago, CoRSU, Katalemwa Cheshire Home) and parent support groups.
- **Every survivor of moderate or severe HIE** needs follow-up for cerebral palsy, epilepsy, developmental delay, hearing and vision problems, and feeding difficulty. Mild HIE still needs a developmental review.

Fluids and nutrition

Day	Typical total fluids in HIE	Notes
Day 1	**60 mL/kg/day** 10% dextrose (some units 40–60 if oliguric)	Glucose infusion rate ≈ 4 mg/kg/min at 60 mL/kg/day
Day 2–3	Increase by 20–30 mL/kg/day **only if** passing urine and sodium normal	Add sodium after 48 h (for example 10% dextrose with 0.18% saline)
Oliguria / AKI	**Insensible losses (30–40 mL/kg/day) + urine output**	No potassium; daily weight and electrolytes

- **Glucose infusion rate (GIR)** = (% dextrose × mL/kg/day) ÷ 144.
- **Feeding**: start **EBM by NGT** (10–20 mL/kg/day) once the baby is stable (no seizures for 24 h, abdomen soft, bowel sounds); advance by 20–30 mL/kg/day; move to cup or breast as suck returns. Severe HIE babies may need long-term NGT feeding.

Temperature and therapeutic hypothermia

- **Therapeutic hypothermia (TH)** = whole-body cooling to **33.5 °C for 72 hours**, started **within 6 hours** of birth, followed by slow rewarming (0.5 °C per hour), for babies **≥36 weeks** with **moderate or severe HIE**, in a **NICU** with ventilation, monitoring and blood gases. In high-income settings it reduces death or disability (number needed to treat about 6–9).
- **HELIX trial** (Lancet Global Health 2021; India, Sri Lanka, Bangladesh): TH did **not** reduce death or disability and **significantly increased death** in LMIC tertiary units. Many babies in LMICs have longer, partial insults, more infection, and different timing, so the benefit from high-income trials does not transfer.

> **UGANDA CONTEXT**
> Therapeutic hypothermia is **not standard care** in Ugandan public hospitals and should be offered only within a research protocol or a fully equipped NICU. On the ward, the priority is the opposite mistake: **avoid hyperthermia** (radiant warmers set too high, heavy wrapping, fever). Keep the baby at **36.5–37.5 °C**.

Other organ support

- **Respiratory**: meconium aspiration → oxygen, CPAP, antibiotics; PPHN (labile SpO_2, pre- and post-ductal difference) → oxygen, avoid acidosis and hypothermia, refer.
- **Cardiovascular**: hypotension → cautious saline (only if hypovolaemic), inotropes (dopamine 5–10 µg/kg/min) where available.
- **Renal**: fluid restriction, avoid nephrotoxins, adjust gentamicin; monitor potassium.
- **Liver and blood**: vitamin K, FFP and platelets for bleeding.
- **Gut**: delay or slow feeds; watch for NEC.

Communicating with the family

- **Breaking news**: be honest and gentle; use plain words ("the baby's brain was short of oxygen during labour"); avoid blame; avoid firm predictions on day 1; review with the family daily; involve the father; screen the mother for depression.

10. Complications

Disease-related

Acute	Long-term
Seizures, cerebral oedema	**Cerebral palsy** (often spastic quadriplegia or dyskinetic)
Apnoea, respiratory failure, PPHN, meconium aspiration	Epilepsy
AKI, SIADH, hyponatraemia	Intellectual disability, developmental delay
Myocardial dysfunction, hypotension	Visual impairment (cortical)
Hypoglycaemia, hypocalcaemia	Hearing loss
DIC, liver injury	Feeding difficulty, aspiration, malnutrition
NEC, feed intolerance	Microcephaly

Acute	Long-term
Sepsis	Behavioural problems, family stress

Treatment-related

Complication	Cause	Recognition and action
Apnoea, hypotension	Stacked anticonvulsants (phenobarbital, diazepam, midazolam)	Bag and mask ready; avoid diazepam
Fluid overload, hyponatraemia, cerebral oedema	"Routine" maintenance fluids in an oliguric baby	Restrict to insensible losses + urine output
Worse brain injury	Hyperthermia (radiant warmer too high), hyperoxia, hypocapnia from over-ventilation	Temperature 36.5–37.5 °C; SpO_2 90–95%; gentle ventilation
Hyperglycaemia	Excess glucose infusion	Check glucose; reduce GIR
Nephrotoxicity, ototoxicity	Gentamicin in AKI	Extend interval; stop if culture negative
Death	Uncontrolled therapeutic hypothermia in LMIC settings (HELIX)	Do not cool outside a NICU protocol
Extravasation burns, bradycardia	Calcium gluconate given fast or into a tissued line	Slow IV with HR monitoring

11. Prognosis

- **Mild HIE**: usually normal outcome; still review development.
- **Moderate HIE**: significant risk of death or disability (roughly a third or more without cooling); outcome depends on recovery.
- **Severe HIE**: high mortality; most survivors have **cerebral palsy** (often spastic quadriplegia or dyskinetic), microcephaly, epilepsy, intellectual disability, cortical visual impairment, hearing loss, feeding problems.
- **Good signs**: normal tone, alertness and **sucking by day 7** (WHO: "an infant who is normally active will usually do well"); seizures controlled early; normal neurological examination at discharge.
- **Bad signs**: severe (Sarnat 3) encephalopathy, prolonged resuscitation (no HR at 10 min), seizures that are hard to control, abnormal examination beyond 1 week, **inability to suck at 1 week**, abnormal EEG or MRI, microcephaly at follow-up. A baby still floppy or spastic, unresponsive and unable to suck at 1 week has severe brain injury and will do poorly (WHO).

12. Prevention

- **Prevention is the treatment.** A well-monitored labour (partograph, fetal heart every 30 min in active labour, timely action) and a trained person with a working bag and mask at every birth prevent most HIE.
- **Primary**: ANC; birth preparedness; facility delivery with a skilled attendant; early referral of prolonged or obstructed labour; careful oxytocin use; timely CS for obstruction, cord prolapse and abruption.
- **At birth**: Helping Babies Breathe (HBB) skills at every birth, with **bag-mask ventilation within the Golden Minute**; a working bag and mask (sizes 0 and 1) checked every shift.
- **Secondary**: avoid secondary injury (normothermia, normoglycaemia, correct fluids, early seizure control, treat infection).
- **Tertiary**: early developmental follow-up and rehabilitation for survivors.

13. Special Populations

- **Preterm babies (<36 weeks)**: encephalopathy has different causes (IVH, periventricular leukomalacia, sepsis); not eligible for cooling; Sarnat staging is less reliable.
- **Babies with AKI**: gentamicin interval must be extended; no potassium in fluids; fluids restricted to insensible losses + urine output.
- **Babies born after instrumental (vacuum) delivery**: look specifically for **subgaleal haemorrhage** (boggy scalp crossing sutures, falling Hb, shock) as a cause or co-existing problem.
- **HIV-exposed babies**: continue PMTCT prophylaxis; give by NGT if unable to suck.

14. Key Points

High-yield points

- Asphyxia is the event; HIE is the brain injury; neonatal encephalopathy is the clinical syndrome.
- Diagnose clinically in Uganda: hypoxic labour + resuscitation + encephalopathy.
- Grade with Sarnat and re-grade daily: mild = hyperalert, no seizures; moderate = lethargy, hypotonia, seizures; severe = coma, flaccid.
- Two waves of injury; treatment prevents secondary injury.
- Temperature 36.5–37.5 °C; **avoid hyperthermia**; no ad hoc cooling (HELIX).
- SpO_2 90–95%; avoid hyperoxia and over-ventilation.
- Glucose ≥2.6 mmol/L; avoid hyperglycaemia.
- 10% dextrose 60 mL/kg/day on day 1; restrict if oliguric (AKI, SIADH).
- Phenobarbital 20 mg/kg, up to 40 mg/kg; maintenance 5 mg/kg/day; no routine diazepam.
- Always exclude sepsis, meningitis and hypoglycaemia.
- Suck and tone at 1 week are the best bedside prognostic guide.
- Follow up every survivor of moderate or severe HIE.

Numbers & doses

Item	Value
Asphyxia supporting criteria	**pH <7.0**, **base excess ≥12**, Apgar **0–3 beyond 5 min**, encephalopathy, multi-organ failure
Latent phase / therapeutic window	About **6 hours**
Cooling (where offered)	**33.5 °C for 72 h**, start within **6 h**, **≥36 weeks**, moderate or severe HIE
HELIX trial (LMIC, 2021)	No benefit; **more deaths** with cooling
Temperature target	**36.5–37.5 °C**; avoid >37.5 °C
SpO_2 target	**90–95%**
Glucose	Treat **<2.6 mmol/L** with **10% dextrose 2 mL/kg**
Day-1 fluids	**10% dextrose 60 mL/kg/day** (restrict if oliguric)
Oliguria	**<1 mL/kg/h**
GIR formula	**(% dextrose × mL/kg/day) ÷ 144**
Phenobarbital	**20 mg/kg** load (+10 mg/kg to **40 mg/kg**); maintenance **5 mg/kg/day**
Phenytoin	**20 mg/kg** IV over 20–30 min in saline

Item	Value
Levetiracetam	**40–60 mg/kg** loading
Calcium gluconate 10%	**2 mL/kg** slow IV
Saline bolus (if hypovolaemic)	**10 mL/kg**
Prognostic checkpoint	Suck and tone at **1 week**

Exam traps

- **"Apgar of 3 at 1 minute = birth asphyxia."** → Apgar alone does not diagnose asphyxia; it is affected by prematurity, drugs and resuscitation.
- **"Cool all asphyxiated babies with a fan and wet towels."** → Wrong. Uncontrolled cooling is dangerous; **avoid hyperthermia** and offer TH only within NICU protocols.
- **"Give diazepam for neonatal seizures."** → Phenobarbital is first line; diazepam causes apnoea and is not recommended routinely.
- **"Give full maintenance fluids from day 1 and increase daily."** → Restrict if oliguric; AKI and SIADH are common.
- **"No seizures = mild HIE."** → Severe HIE may have few clinical seizures; grade on **consciousness and tone**.
- **"Floppy baby after a long labour: it's HIE; no need for antibiotics."** → Always exclude **sepsis, meningitis and hypoglycaemia**.
- **"Mild HIE needs no follow-up."** → Review development; mild cases can still have subtle problems.

Questions seniors ask

Viva questions

1. *What is the difference between birth asphyxia and HIE?* Asphyxia is the event (hypoxia, hypercapnia, acidosis, failure to breathe); HIE is the brain injury that results.
2. *Describe Sarnat stage 2.* Lethargy, hypotonia, weak suck and Moro, constricted pupils, bradycardia, seizures common.
3. *Why avoid hyperthermia?* It increases cerebral metabolic demand and worsens secondary energy failure; even a 1 °C rise increases the odds of death or disability.
4. *First-line anticonvulsant and dose?* Phenobarbital 20 mg/kg IV over 15–20 min; further 10 mg/kg to 40 mg/kg; maintenance 5 mg/kg/day.
5. *Why restrict fluids in HIE?* Risk of AKI and SIADH → fluid overload, hyponatraemia, cerebral oedema.
6. *Is cooling recommended in Uganda?* Not as routine care; HELIX showed increased mortality in LMICs; use only within protocols in fully equipped NICUs.
7. *What predicts a poor outcome?* Severe encephalopathy, prolonged resuscitation (no HR at 10 min), seizures hard to control, abnormal examination and inability to suck at 1 week, abnormal EEG or MRI.

On the ward round

1. **"What Sarnat stage is he, and why?"** Name each feature (consciousness, tone, suck, pupils, seizures).
2. **"How long was he ventilated and when did the heart rate come up?"** Reflects the depth of the insult.
3. **"Why is he on antibiotics?"** Infection risk (ROM, fever) and sepsis cannot be excluded; stop at 48 h if well and culture negative.
4. **"What is his urine output and fluid plan?"** mL/kg/h and total fluids mL/kg/day.
5. **"What anticonvulsant and how much?"** Phenobarbital in mg/kg and total mg.
6. **"Would you cool this baby?"** Not in our setting outside a NICU protocol (HELIX); keep normothermic.
7. **"What will you tell the parents about the future?"** Honest uncertainty; review suck and tone over the week; follow-up for development, hearing and vision.

Mnemonics

- **Avoid secondary injury: "Keep it in the middle"**: not too hot, not too cold; not too much glucose, not too little; not too much oxygen, not too little; not too much fluid, not too little.
- **Multi-organ injury: "Brain Kills Heart, Lungs, Liver, Gut, Blood"** (brain, kidneys, heart, lungs, liver, gut, blood/DIC).
- **Sarnat: "1 = Hyper, 2 = Hypo + fits, 3 = Coma + flat"**.

OSCE tips

- Expect a station on **Sarnat staging** from a video or description, or on **calculating phenobarbital doses**.
- When counselling parents, use plain words ("the baby's brain was short of oxygen during labour"), avoid blame, explain uncertainty, describe the plan and follow-up.
- In an examination station, record each Sarnat item aloud, state the stage, and give the time.

15. References

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	WHO definition of birth asphyxia.	**Failure to initiate and sustain breathing at birth.** Physiologically: impaired gas exchange causing hypoxaemia, hypercapnia and metabolic acidosis.
2	Difference between neonatal encephalopathy and HIE.	**Neonatal encephalopathy** is the clinical syndrome (altered consciousness, tone, feeding, seizures) in a baby of 35 weeks or more; **HIE** is encephalopathy caused by intrapartum hypoxia-ischaemia.
3	Criteria supporting an intrapartum hypoxic-ischaemic cause.	Cord/early arterial **pH below 7.0** and/or **base deficit 12 mmol/L or more**; **Apgar 0-3 beyond 5 min** (or below 5 at 5 and 10 min); early moderate/severe encephalopathy; multi-organ dysfunction; sentinel event; compatible imaging.
4	Sarnat stage 1 (mild) features.	**Hyperalert**, irritable, jittery; normal or increased tone; strong Moro; weak suck; **dilated pupils** (sympathetic); **no seizures**; lasts under 24 h; usually **normal outcome**.
5	Sarnat stage 2 (moderate) features.	**Lethargic**, **mild hypotonia**, weak Moro and suck, strong distal flexion, **constricted pupils**, bradycardia (parasympathetic), **seizures common**; lasts 2-14 days; variable outcome.
6	Sarnat stage 3 (severe) features.	**Stupor/coma**, **flaccid**, absent reflexes (Moro, suck), decerebrate posturing, variable/poorly reactive pupils, apnoea; seizures may be suppressed; **high mortality, survivors usually disabled**.
7	What happens in the latent phase of HIE and why does it matter?	After resuscitation, brain energy partly recovers for about **6 hours** before **secondary energy failure** (6-72 h). This is the **therapeutic window**: cooling must start within 6 h; supportive care aims to prevent secondary injury.
8	Name six factors that worsen secondary brain injury.	**Hyperthermia**, **hypoglycaemia** (and hyperglycaemia), **seizures**, **hypotension**, hypoxaemia/**hyperoxia**, **hypocapnia** (over-ventilation), infection, fluid overload.
9	Target temperature for a baby with HIE in a Ugandan ward.	**36.5-37.5 C**. Avoid **hyperthermia** (radiant warmer too high, overwrapping). Do not actively cool outside a NICU protocol.
10	Therapeutic hypothermia protocol (where offered).	Whole-body cooling to **33.5 C for 72 h**, started **within 6 h** of birth, for babies **36 weeks or more** with **moderate/severe HIE**, then rewarm at 0.5 C/h; needs NICU support.
11	What did the HELIX trial show?	In tertiary units in **India, Sri Lanka and Bangladesh** (Lancet Global Health 2021), cooling **did not reduce death or disability** and **significantly increased death**. So cooling is not standard care in LMICs like Uganda.
12	Phenobarbital regimen for neonatal seizures.	**Loading 20 mg/kg IV** over 15-20 min; if seizures persist, further **10 mg/kg** doses to **40 mg/kg total**; maintenance **5 mg/kg/day once daily**. Watch for apnoea.
13	Second-line anticonvulsants in neonates.	**Phenytoin 20 mg/kg IV** over 20-30 min in saline (cardiac monitoring), or **levetiracetam 40-60 mg/kg**, or midazolam (NICU). Diazepam is not routine (apnoea, displaces bilirubin).
14	Day-1 fluids in HIE.	**10% dextrose 60 mL/kg/day**, restricted (e.g. 40 mL/kg/day or insensible losses + urine output) if **oliguric or hyponatraemic**.

#	Question	Answer
15	Define oliguria in a neonate.	Urine output **below 1 mL/kg/h** (after the first 24 h). Weigh nappies: 1 g = 1 mL.
16	Glucose infusion rate (GIR) formula.	**GIR (mg/kg/min) = (% dextrose × mL/kg/day) ÷ 144.** 10% at 60 mL/kg/day = 4.2 mg/kg/min.
17	Why does asphyxia injure the kidneys and gut?	The **diving reflex** redistributes blood to the **brain, heart and adrenals**, starving the kidneys, gut, liver, skin and lungs → AKI, NEC/feed intolerance, liver injury.
18	Treatment of hypocalcaemic seizures in a neonate.	**10% calcium gluconate 2 mL/kg slow IV** over about 10 min with heart rate monitoring (risk of bradycardia and extravasation burns), then maintenance.
19	WHO clinical prognostic rule for HIE.	A baby who **within a week is still floppy or spastic, unresponsive and cannot suck** has severe brain injury and will do poorly; one who is normally active usually does well.
20	Main long-term complications of HIE.	**Cerebral palsy** (spastic quadriplegia, dyskinetic), **epilepsy**, intellectual disability, cortical visual impairment, hearing loss, microcephaly, feeding difficulty.
21	Basal ganglia/thalamus vs watershed injury pattern.	**Acute profound** insult (cord prolapse, rupture, abruption) → basal ganglia/thalami → dyskinetic/spastic CP. **Prolonged partial** insult (long obstructed labour) → watershed cortex/white matter → cognitive problems, spastic CP.
22	Pupils in mild vs moderate HIE.	Mild: **dilated** (sympathetic overactivity). Moderate: **constricted** (parasympathetic), with bradycardia and secretions.
23	What should a HC III/IV do for an asphyxiated baby before referral?	HBB resuscitation; keep warm (not hot); glucose check or EBM/sugar water by cup/NGT; **phenobarbital 20 mg/kg** (IV/IM) if convulsing; first dose of **ampicillin + gentamicin** if risk factors; refer with the mother and a note.
24	Important mimics of HIE to exclude.	**Sepsis/meningitis**, **hypoglycaemia**, intracranial or **subgaleal haemorrhage**, maternal drugs (opioids, MgSO4), hypocalcaemia, inborn errors of metabolism, neonatal stroke, neuromuscular disease.
25	When do HIE seizures usually start?	Usually **within the first 12-24 hours** of life; often **subtle** (lip smacking, eye deviation, cycling, apnoea).
26	Thompson score.	A 0-22 bedside HIE score (tone, consciousness, fits, posture, Moro, grasp, suck, respiration, fontanelle) used in LMIC research and some Ugandan units; higher = worse; about 15 or more predicts poor outcome.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A term baby born after a 3-hour second stage needed 4 minutes of bag-mask ventilation. At 12 hours he is lethargic, hypotonic, with a weak suck, constricted pupils and an episode of lip smacking with eye deviation. Which Sarnat stage is he?

A Stage 1 (mild)
B Normal post-resuscitation state
C Stage 2 (moderate)
D Stage 3 (severe)
E Cannot be staged without an EEG

2. A 3.0 kg term baby with moderate HIE has a generalised clonic seizure at 8 hours. Glucose is 4.1 mmol/L. IV access is in place. What is the MOST appropriate first drug?

A Phenobarbital 60 mg IV over 15-20 minutes
B Diazepam 3 mg IV repeated every 5 minutes
C Phenytoin 60 mg IV push over 1 minute
D 10% dextrose 6 mL IV
E Paracetamol 45 mg rectally

3. A baby with moderate HIE is under a radiant warmer on day 1. The nurse records an axillary temperature of 38.2 C. There are no other signs of infection. What is the MOST appropriate immediate action?

A Increase the radiant warmer to prevent shivering
B Give paracetamol 15 mg/kg and continue as before
C Apply ice packs and a fan to bring the temperature to 33.5 C
D Reduce/turn off the warmer, remove extra wraps, recheck in 15-30 min, and consider sepsis
E Ignore it; hyperthermia is protective after asphyxia

4. A 3.2 kg baby with moderate HIE on 10% dextrose 60 mL/kg/day has passed 15 mL of urine in 24 hours. Weight is now 3.35 kg, sodium 127 mmol/L, and he has mild periorbital oedema. What is the MOST appropriate fluid plan?

A Increase fluids to 90 mL/kg/day as per day-2 routine
B Give furosemide and increase fluids to 120 mL/kg/day
C Give normal saline 20 mL/kg bolus to improve kidney perfusion
D Add potassium chloride to the infusion
E Restrict fluids to insensible losses plus urine output, avoid potassium, recheck electrolytes and creatinine

5. Which statement about therapeutic hypothermia for HIE is MOST accurate for a Ugandan regional referral hospital?

A It should be started at any time in the first week
B It is not standard care; in LMIC units the HELIX trial found no benefit and more deaths, so the priority is to avoid hyperthermia
C It is first-line care for all asphyxiated babies, including preterm babies below 34 weeks
D Passive cooling with fans is a safe alternative when no device is available
E It cures HIE if started within 24 hours

6. A term baby is floppy and lethargic at 30 hours. Labour was normal and short; Apgar 9 and 10. The mother had a fever of 38.6 C in labour and membranes ruptured 22 hours before birth. The fontanelle is full. What is the MOST likely diagnosis?

A Mild HIE
B Severe HIE
C Neonatal sepsis with possible meningitis
D Maternal opioid effect
E Benign neonatal sleep myoclonus

7. A baby born by vacuum extraction after a prolonged second stage becomes pale and tachycardic at 4 hours. There is a boggy, fluctuant scalp swelling crossing the suture lines, extending to the neck, and the head circumference has increased by 2 cm. What is the MOST appropriate next step?

A Aspirate the swelling to reduce pressure
B Reassure; it is caput succedaneum and will resolve in 48 hours
C Apply a tight head bandage and continue routine care
D Start phenobarbital for presumed HIE
E Treat as subgaleal haemorrhage: IV access, saline 10 mL/kg and urgent blood, vitamin K, check Hb and clotting, monitor head circumference

8. A 3.5 kg baby with HIE continues to have seizures 30 minutes after a 70 mg phenobarbital loading dose. Glucose is 3.8 mmol/L. What is the MOST appropriate next step?

A Give another phenobarbital dose of 35 mg (10 mg/kg) IV slowly, with a bag and mask ready
B Give a second full 70 mg dose immediately as an IV push
C Give diazepam 5 mg IV
D Wait 24 hours for the phenobarbital to work
E Give 10% dextrose 2 mL/kg

9. Which finding at 7 days of life BEST predicts a poor neurological outcome in a term baby with HIE, when MRI and EEG are not available?

A A single seizure on day 1 that was controlled with phenobarbital
B Physiological jaundice on day 4
C Still unresponsive, floppy or spastic, and unable to suck
D Weight loss of 7% from birth weight
E Apgar 4 at 1 minute

10. After asphyxia, blood is redirected to protect certain organs. Which organ is MOST protected by this diving reflex?

A Kidneys
B Gut
C Skin
D Adrenal glands
E Liver

11. A term baby at a HC IV needed 3 minutes of bag-mask ventilation, now breathes at 50/min but is lethargic with a weak suck. The nearest SCBU is 2 hours away. Which plan is MOST appropriate before referral?

A Discharge home with the mother; return if seizures

B Give expressed colostrum by cup/NGT or check glucose, keep normothermic (skin-to-skin), and refer with a referral note and the mother

C Wrap in three blankets under a heater because asphyxiated babies get cold

D Give phenobarbital 20 mg/kg as prophylaxis even though there are no seizures

E Give sodium bicarbonate to correct presumed acidosis

12. Which pattern of brain injury is MOST typical after an acute, near-total insult such as a cord prolapse with 25 minutes of profound bradycardia?

A Parasagittal watershed injury only

B Periventricular leukomalacia

C Isolated germinal matrix haemorrhage

D Basal ganglia and thalamic injury

E Cerebellar hypoplasia

13. A baby with moderate HIE on day 2 has a seizure. Glucose 3.9 mmol/L, total calcium 1.5 mmol/L (low). He has already received 40 mg/kg phenobarbital. What is the MOST appropriate next step?

A 10% calcium gluconate 2 mL/kg slow IV over 10 minutes with heart-rate monitoring

B Another 20 mg/kg phenobarbital

C 10% calcium gluconate 2 mL/kg as a rapid IV push

D Oral calcium only

E Magnesium sulphate 50% 1 mL/kg IV push

14. When is it MOST appropriate to start enteral feeds in a baby with moderate HIE?

A Immediately at birth, full volume by bottle

B Only after discharge

C After 2 weeks of IV fluids

D When stable: no seizures for about 24 hours, abdomen soft with bowel sounds; start small EBM feeds by NGT and advance as tolerated

E Only when the baby can suck at the breast

15. Which statement about the Apgar score is CORRECT?

A A 1-minute Apgar of 3 confirms birth asphyxia

B Resuscitation should start after the 1-minute Apgar is recorded

C The Apgar score describes the baby's condition but alone does not diagnose asphyxia; an Apgar of 0-3 beyond 5 minutes supports a hypoxic cause

D A 5-minute Apgar of 7 rules out HIE

E Apgar scores are not affected by prematurity or maternal drugs

Answers & explanations

> **1. C** · Stage 2 (moderate)
>
> C is right: lethargy, hypotonia, weak suck, constricted pupils (parasympathetic) and seizures (the lip smacking and eye deviation are subtle seizures) define Sarnat stage 2. A is wrong: stage 1 babies are hyperalert with normal or increased tone, dilated pupils and no seizures. B is wrong: encephalopathy with seizures is not normal. D is wrong: stage 3 babies are stuporous/comatose and flaccid with absent reflexes. E is wrong: Sarnat staging is clinical; EEG adds detail but is not required.

2. A · Phenobarbital 60 mg IV over 15-20 minutes

A is right: phenobarbital is first line (WHO/UCG); 20 mg/kg × 3.0 kg = 60 mg IV slowly over 15-20 min, with extra 10 mg/kg doses to 40 mg/kg if needed. B is wrong: diazepam is not recommended routinely in neonates (apnoea) and 1 mg/kg repeated is an overdose. C is wrong: phenytoin is second line and must be given slowly (max 1 mg/kg/min) with cardiac monitoring. D is wrong: glucose is normal, so dextrose will not stop the seizure. E is wrong: there is no fever, and paracetamol is not an anticonvulsant.

3. D · Reduce/turn off the warmer, remove extra wraps, recheck in 15-30 min, and consider sepsis

D is right: hyperthermia worsens secondary brain injury after asphyxia; the commonest cause on the ward is environmental (warmer/wraps). Correct it and recheck, while keeping sepsis in mind. A is wrong: it makes hyperthermia worse. B is wrong: antipyretics are not used in neonates (WHO: control the environment) and do not address the cause. C is wrong: uncontrolled cooling with ice and fans outside a NICU protocol is dangerous, and HELIX showed increased mortality with cooling in LMICs. E is wrong: hyperthermia is harmful, not protective.

4. E · Restrict fluids to insensible losses plus urine output, avoid potassium, recheck electrolytes and creatinine

E is right: urine output is about 0.2 mL/kg/h (oliguria), with weight gain, oedema and hyponatraemia: AKI and/or SIADH with fluid retention. Restrict to insensible losses (about 30-40 mL/kg/day) plus urine output, avoid potassium, monitor creatinine and sodium. A is wrong: routine increases cause fluid overload and cerebral oedema. B is wrong: increasing fluids is the opposite of what is needed. C is wrong: he is overloaded, not hypovolaemic. D is wrong: potassium is dangerous in AKI (hyperkalaemia).

5. B · It is not standard care; in LMIC units the HELIX trial found no benefit and more deaths, so the priority is to avoid hyperthermia

B is right: HELIX (India, Sri Lanka, Bangladesh, 2021) found that cooling did not reduce death or disability and increased mortality; cooling is offered only within NICU protocols, and the ward priority is normothermia (36.5-37.5 C). A is wrong: the window is within 6 hours. C is wrong: cooling trials were in babies of 36 weeks or more with moderate/severe HIE. D is wrong: uncontrolled cooling causes overcooling and harm. E is wrong: even in HICs cooling reduces, not abolishes, death/disability (NNT about 6-9), and it must start within 6 h.

6. C · Neonatal sepsis with possible meningitis

C is right: a normal labour with good Apgar scores makes HIE unlikely; maternal fever, prolonged ROM and a full fontanelle point to early-onset sepsis/meningitis. Take cultures, do an LP when stable, and start ampicillin + gentamicin (meningitis doses). A and B are wrong: HIE requires a hypoxic insult and is present from birth; this baby was vigorous. D is wrong: opioid depression is present at birth and wears off over hours, not starting at 30 h. E is wrong: benign sleep myoclonus occurs in a well, alert baby only during sleep.

7. E · Treat as subgaleal haemorrhage: IV access, saline 10 mL/kg and urgent blood, vitamin K, check Hb and clotting, monitor head circumference

E is right: a fluctuant swelling crossing sutures after vacuum, with pallor, tachycardia and rising head circumference, is a subgaleal haemorrhage, which can hold most of a baby's blood volume. Resuscitate with volume and blood, give vitamin K and correct coagulopathy. A is wrong: aspiration introduces infection and does not stop bleeding. B is wrong: caput is a soft oedematous swelling present at birth without shock. C is wrong: bandaging does not control the bleed and delays resuscitation. D is wrong: the problem is haemorrhagic shock, not HIE seizures.

8. A · Give another phenobarbital dose of 35 mg (10 mg/kg) IV slowly, with a bag and mask ready

A is right: if seizures persist after the 20 mg/kg load, WHO advises further 10 mg/kg doses up to 40 mg/kg total (here 35 mg slowly), watching for apnoea. B is wrong: a second 20 mg/kg as a push risks apnoea and hypotension; doses should be given slowly in 10 mg/kg steps. C is wrong: 5 mg diazepam (about 1.4 mg/kg) is an overdose and diazepam is not routine in neonates. D is wrong: ongoing seizures worsen brain injury and must be treated. E is wrong: glucose is normal.

9. C · Still unresponsive, floppy or spastic, and unable to suck

C is right: the WHO Pocket Book states that an infant who within a week is still floppy or spastic, unresponsive and cannot suck has severe brain injury and will do poorly. A is wrong: a single early, controlled seizure is less predictive. B is wrong: physiological jaundice is unrelated. D is wrong: 7% weight loss is within normal limits. E is wrong: the 1-minute Apgar has poor prognostic value; prolonged low Apgar (e.g. 0-3 at 10 min) is more meaningful.

10. D · Adrenal glands

D is right: during hypoxia, blood is redistributed to the brain, heart and adrenals. A, B, C and E are wrong: the kidneys (AKI), gut (NEC, feed intolerance), skin and liver (raised enzymes, coagulopathy, hypoglycaemia) are sacrificed, which explains multi-organ dysfunction after asphyxia.

11. B · Give expressed colostrum by cup/NGT or check glucose, keep normothermic (skin-to-skin), and refer with a referral note and the mother

B is right: an encephalopathic baby needs SCBU care; before referral, prevent hypoglycaemia (EBM or check glucose), keep 36.5-37.5 C (skin-to-skin), give antibiotics if risk factors, and refer with documentation. A is wrong: a lethargic baby with a weak suck after resuscitation must not go home. C is wrong: overheating worsens brain injury. D is wrong: prophylactic phenobarbital is not recommended without seizures. E is wrong: sodium bicarbonate is listed by UCG as a harmful practice in newborn resuscitation.

12. D · Basal ganglia and thalamic injury

D is right: an acute profound insult damages the most metabolically active deep grey matter (basal ganglia and thalami), leading to dyskinetic or spastic quadriplegic CP. A is wrong: watershed injury follows prolonged partial insults. B and C are wrong: PVL and germinal matrix haemorrhage are injuries of preterm babies. E is wrong: cerebellar hypoplasia is developmental, not an acute hypoxic pattern.

13. A · 10% calcium gluconate 2 mL/kg slow IV over 10 minutes with heart-rate monitoring

A is right: hypocalcaemia is a treatable cause of seizures after asphyxia; WHO advises 10% calcium gluconate 2 mL/kg as a slow IV infusion, monitoring heart rate, then supplementation. B is wrong: he has reached the 40 mg/kg maximum and the cause is metabolic. C is wrong: a rapid push causes bradycardia/arrhythmia and extravasation burns. D is wrong: oral calcium is too slow for an acute seizure. E is wrong: magnesium may be needed if hypomagnesaemic, but 50% undiluted 1 mL/kg as a push is dangerous and not first line here.

14. D · When stable: no seizures for about 24 hours, abdomen soft with bowel sounds; start small EBM feeds by NGT and advance as tolerated

D is right: once stable, start small volumes of expressed breast milk by NGT (e.g. 10-20 mL/kg/day) and advance by 20-30 mL/kg/day, moving to cup/breast as the suck returns. A is wrong: an encephalopathic baby cannot coordinate feeds safely, and the gut may be ischaemic. B and C are wrong: prolonged IV fluids increase infection risk and delay recovery. E is wrong: waiting for a good suck may take days to weeks; NGT feeding allows nutrition meanwhile.

15. C · The Apgar score describes the baby's condition but alone does not diagnose asphyxia; an Apgar of 0-3 beyond 5 minutes supports a hypoxic cause

C is right: Apgar is a description that is affected by many factors; prolonged low scores support (but do not prove) an intrapartum hypoxic cause. A is wrong: a single low 1-minute score is common and often recovers. B is wrong: resuscitation must start within the Golden Minute, never waiting for the Apgar. D is wrong: some babies with HIE have reasonable 5-minute scores, especially after effective resuscitation. E is wrong: prematurity, maternal sedation and congenital anomalies lower Apgar scores.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Obstructed labour, a non-breathing baby and seizures

A 19-year-old primigravida is referred from a HC III in Wakiso to Mengo after 16 hours of labour with no progress. She has an emergency CS for obstructed labour. The baby boy is born with thick meconium, floppy, not breathing, HR 60/min. He is ventilated with bag and mask; HR rises above 100 at 4 minutes and he gasps at 6 minutes. Apgar 1, 3, 6. Birth weight 3.3 kg.

1. After resuscitation, what are your immediate priorities on arrival in the newborn unit?

> Airway and breathing (SpO2 90-95%, avoid hyperoxia; CPAP/bag-mask if apnoea), **temperature 36.5-37.5 C** (avoid overheating), **glucose** now, IV access with **10% dextrose 60 mL/kg/day** (≈ 8.3 mL/h), check perfusion; observe for seizures; vitamin K; inform the senior and the mother.

2. At 6 hours he is lethargic, hypotonic, weak suck, pupils constricted, HR 105. Stage him and list what you will monitor.

> **Sarnat stage 2 (moderate HIE).** Monitor 1-2-hourly: temperature, HR, RR/apnoea, SpO2, seizures; glucose 3-6-hourly; urine output (weigh nappies), daily weight, electrolytes/creatinine where available; repeat Sarnat at least daily.

3. At 10 hours he has cycling movements of the legs and eye deviation for 2 minutes. What will you do (with doses)?

> Airway, oxygen, **check glucose** (and calcium). Give **phenobarbital 20 mg/kg = 66 mg IV over 15-20 min**; if seizures continue, **10 mg/kg = 33 mg** up to a total of **40 mg/kg (132 mg)**; maintenance **5 mg/kg/day = 16.5 mg once daily** from 12-24 h. Bag and mask ready for apnoea.

4. What else could be causing or contributing to his encephalopathy, and how will you check?

> **Sepsis/meningitis** (long labour, multiple vaginal examinations at HC III: blood culture, CRP, LP when stable; start ampicillin + gentamicin), **hypoglycaemia** (glucose), **hypocalcaemia** (calcium), **intracranial or subgaleal haemorrhage** (head exam, Hb), meconium aspiration with hypoxaemia (CXR, SpO2).

5. On day 2 urine output is 0.4 mL/kg/h and sodium 128 mmol/L. How do you adjust his care?

> **AKI/SIADH**: restrict fluids to insensible losses + urine output (about 30-40 mL/kg/day + measured output), no potassium, recheck electrolytes and creatinine, **extend gentamicin interval or stop it** if cultures negative at 48 h, weigh twice daily.

6. On day 7 he is alert, tone improving and sucking from a cup. How do you counsel the family about prognosis and follow-up?

> Improvement in alertness, tone and suck by day 7 is **encouraging**, but moderate HIE carries a risk of CP, epilepsy and developmental delay. Plan: continue breastfeeding/cup feeds, review at 6 weeks and then 3, 6, 9, 12 and 18 months for development, head growth, hearing and vision; early physiotherapy if tone abnormal; decide on stopping phenobarbital (senior); immunise (BCG/OPV0). Be honest and supportive; give danger signs.

7. What could have prevented this?

> Partograph use and **early referral** when the alert/action lines were crossed, timely CS, fetal heart monitoring, avoiding long delays in transport, and skilled resuscitation at birth. Discuss in the perinatal death/near-miss (MPDSR) review without blame.

Take-away: In moderate HIE, keep the baby normothermic, glycaemic and well ventilated, treat seizures with phenobarbital in mg/kg, restrict fluids when urine output falls, exclude sepsis, and judge prognosis by recovery of suck and tone over the first week.

Case 2: A floppy baby after vacuum delivery

At Kawempe National Referral Hospital, a 4.0 kg baby girl was born by vacuum extraction after a 2.5-hour second stage. She cried at 2 minutes after stimulation (Apgar 5 and 8). At 3 hours the midwife calls you because she is pale and 'not active'. HR 182/min, RR 64/min, CRT 4 seconds, temperature 36.8 C. Glucose 3.2 mmol/L.

1. What examination finding are you specifically looking for, and why?

> A **boggy, fluctuant scalp swelling crossing suture lines**, spreading to the forehead/neck, with **increasing head circumference** = **subgaleal haemorrhage** (vacuum), which can cause hypovolaemic shock. Compare with caput (soft oedema, crosses sutures, present at birth, no shock) and cephalhaematoma (bounded by sutures, no shock).

2. You find a fluctuant swelling over the whole scalp. What is your immediate management?

> **Call for help**; IV/IO access; **normal saline 10 mL/kg (40 mL)** over 5-10 min, repeat as needed; urgent **cross-match**, give **O-negative/group-specific blood** 10-20 mL/kg; **vitamin K**; check Hb, platelets, clotting (FFP if DIC); monitor HR, CRT, head circumference hourly; keep warm; nil by mouth initially.

3. How do you distinguish encephalopathy due to shock from HIE?

> HIE: hypoxic insult before/at birth, needs resuscitation, encephalopathy from birth. This baby was relatively vigorous (Apgar 8 at 5 min) and deteriorated later with **signs of shock** (tachycardia, prolonged CRT, pallor): **hypovolaemia** from bleeding is the main cause, though shock can cause secondary hypoxic-ischaemic injury.

4. What complications will you monitor for over the next days?

> **Hypovolaemic shock**, DIC, **jaundice** (breakdown of the haematoma: check bilirubin), seizures, AKI, infection. Monitor Hb, bilirubin, urine output, neurology.

5. How do you counsel the parents?

> Explain that bleeding happened under the scalp after the vacuum delivery, what you are doing (fluids, blood), that the swelling resolves over weeks, the risk of jaundice and the need for observation, and follow-up for development. Document and report through clinical audit.

Take-away: Not every floppy baby after a difficult delivery has HIE: a pale, tachycardic baby after vacuum delivery may be bleeding into a subgaleal haemorrhage and needs volume and blood immediately.

Case 3: Day 5: is this still asphyxia?

A term 3.1 kg baby boy admitted with mild HIE (hyperalert, jittery, no seizures) after a prolonged labour is ready for discharge on day 3. On day 5 the mother returns to Mengo because he has stopped feeding and is 'too sleepy'. Temperature 38.3 C, HR 176, RR 66, mild chest indrawing, fontanelle full. Glucose 2.4 mmol/L.

1. What is the most likely diagnosis and why is it not worsening HIE?

> **Late-onset neonatal sepsis, possibly meningitis.** HIE is maximal in the first 72 h and then improves; a baby who was improving and then deteriorates with **fever, tachycardia, tachypnoea and a full fontanelle** has a new problem.

2. What are your first actions?

> ABC; oxygen if SpO2 below 90%; **10% dextrose 2 mL/kg = 6 mL IV** and recheck in 30 min; maintenance glucose; blood culture; start antibiotics **without delay**; LP when stable (within 2 h of antibiotics ideally).

3. Which antibiotics and doses (day 5, 3.1 kg)?

> Meningitis doses: **ampicillin 100 mg/kg = 310 mg IV 12-hourly** (first week; 8-hourly after day 7) + **gentamicin 5 mg/kg = 15.5 mg IV once daily**; or per UCG/WHO, **ceftriaxone 50 mg/kg 12-hourly or 100 mg/kg once daily** with gentamicin if meningitis likely (avoid ceftriaxone if jaundiced or with IV calcium). Duration **21 days** if meningitis confirmed.

4. What supportive care does he need?

> Temperature control (undress, no paracetamol in neonates per UCG), fluids 10% dextrose with careful volume, NGT EBM when stable, monitor glucose, seizures (phenobarbital if needed), SpO2, urine output.

5. What is the lesson for follow-up of babies with HIE?

> Discharge counselling must include **danger signs** (not feeding, fever, fast breathing, convulsions, sleepiness), and any deterioration after initial improvement should be treated as a **new illness**, most often sepsis.

Take-away: HIE improves after the first 72 hours; a new deterioration with fever or a full fontanelle is sepsis or meningitis until proven otherwise.

References

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.